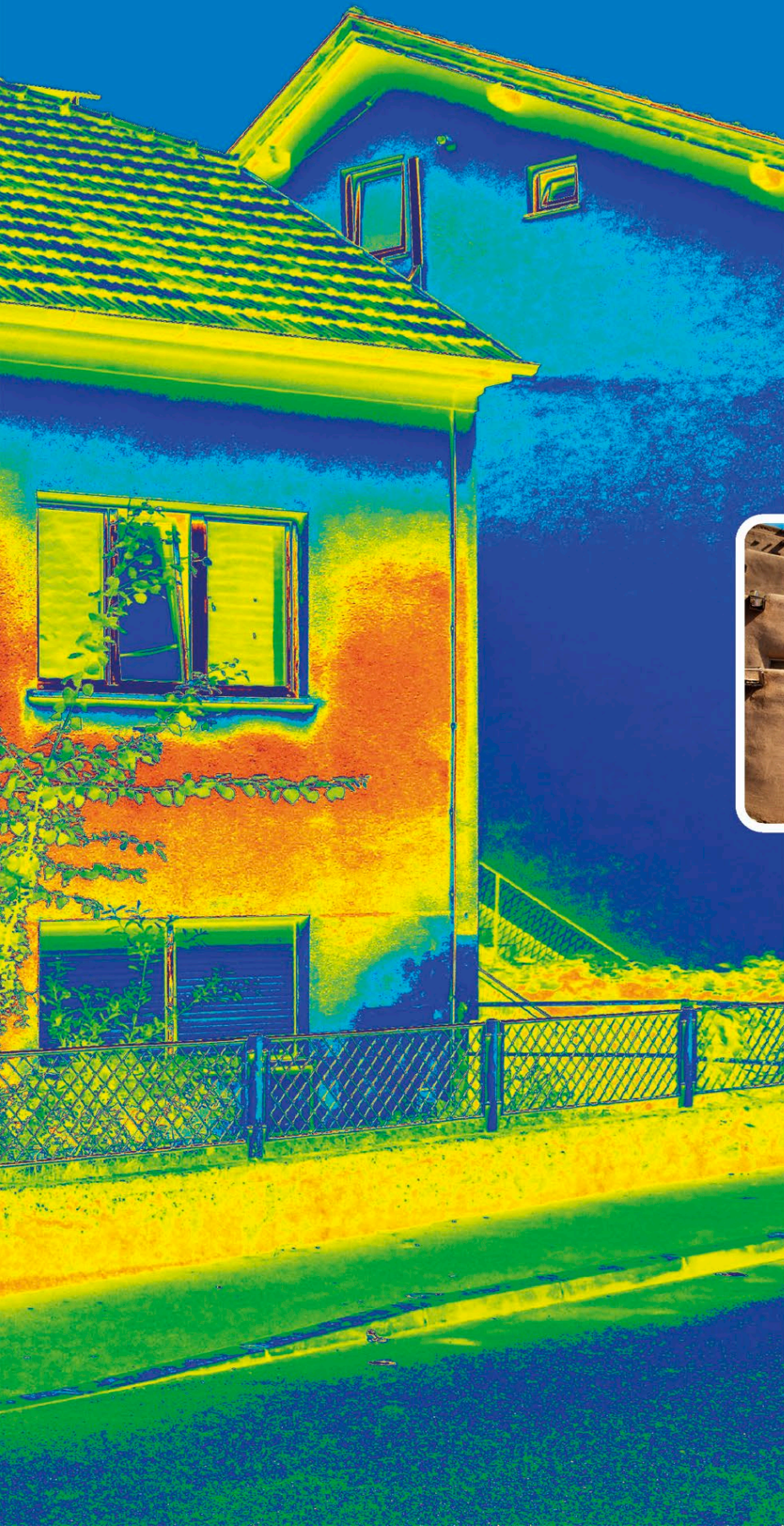


TRAPPING HEAT

HOUSE INSULATION

Heat always moves from warmer to cooler places.

It is easily lost from our homes to the outside air through roofs, walls, doors, and windows. Insulation is a way of reducing the rate of heat lost from the warmer insides of a house to its cooler surroundings during cold winters. In the same way, insulation keeps the house cool in the summer. The roof and walls of modern homes often include a layer of materials through which heat passes slowly, such as foam, fiberglass, and plastic, making them more energy-efficient.



KEEPING COOL

In the hot climate of New Mexico, these homes were designed to stay cool on the inside by using adobe bricks—one of the oldest building materials in the world. They are made of clay mixed with straw or dung.

HEAT TRANSFER

Heat naturally flows from a hot object or area to cooler objects or surroundings. It is transferred in three different ways. Some objects give off heat as waves directly into the air, which is called radiation. Heat can also travel through solids by conduction and through liquids and gases through a process called convection. When water is heated in a saucepan, the heat travels through it in convection currents.

The cooler water sinks to be heated up by convection.

The stove's flame radiates heat in the air.

The warm water becomes less dense and rises.

Heat passes through the pan to the handle by conduction.

